Guideline Medicine

For centuries, treatment decisions in medicine rested on the experience and authority of experts, often with disastrous consequences for patients: bloodletting, prolonged bed rest and milk diets for stomach ulcers, and mutilating surgery for cancer. In the 1970s, a movement emerged at McMaster University in Canada, led by David Sackett, which argued that scientific evidence should be the basis for medical decisions and which began developing the tools and methods to generate and appraise that evidence. Today we call this evidence-based medicine. David Sackett knew very well that for most medical decisions there is no scientific evidence available at all, and that evidence from large clinical trials does not apply to every patient, because such trials always involve a selection of patients. He therefore wrote: “External clinical evidence can inform, but can never replace, individual clinical expertise, and it is this expertise that decides whether the external evidence applies to the individual patient at all and, if so, how it should be integrated into a clinical decision.” A prerequisite for practicing evidence-based medicine, then, is that physicians are able to independently interpret the results of scientific research and apply them to the situation in which a patient presents. In doing so, it is entirely possible that the outcome of a large trial is not followed, or that a treatment is started for which no scientific evidence exists (yet). Sackett even proposed a solution for this: the “N-of-1 trial,” a study in a single patient.

In the Netherlands, evidence-based medicine was enthusiastically embraced from the 1980s onward. Over the years, however, several critical elements of evidence-based medicine have been lost, with highly detrimental effects on current medical practice. We are talking about guideline medicine.

In recent years, many countries have developed medical treatment guidelines; in the Netherlands, this task has been assigned to the professional associations. There are now some 650 guidelines for medical specialists and 150 for general practitioners. A multidisciplinary group of experts formulates recommendations, which are then authorized by the professional associations. This system is time-consuming and expensive, and it has spawned a veritable meeting industry: if roughly 160 guidelines are revised each year, with an average of six experts each spending about twenty-four hours, that adds up to tens of thousands of expert hours per year. Add support, coordination and implementation, and the total cost quickly runs into many millions of euros per year. After three years, 20% of the recommendations are “officially” outdated, but the scientific data on which the guidelines are based are often already outdated by the time a guideline is adopted. This is true for rapidly evolving fields such as cancer and chronic inflammatory diseases, but increasingly for other diseases as well.

Beyond these operational shortcomings, there is a more fundamental problem with guideline medicine. Guidelines are increasingly used as the primary teaching material for medical specialists and general practitioners in training, at the expense of a thorough understanding of the pathophysiology of disease and the mechanisms of action of drugs. We run the risk that physicians mainly learn which recommendation to follow, rather than why that recommendation applies. And once trained, doctors in daily

practice are reluctant to deviate from a guideline, because doing so can lead to legal complications.
Guideline medicine has also led to a flood of useless scientific research in which the guideline itself is effectively the starting point. This accounts for a growing share of medical PhD theses and studies with titles such as “Evaluation of the implementation of guideline…” or “Evaluation of experiences with guideline…”. Such work may be useful for implementing care, but it is no substitute for scientific research into new causes, treatments or mechanisms of action, and it adds no new knowledge.

Artificial intelligence can substantially accelerate the gathering, organizing and flagging of new knowledge, and that should be the basis for drafting guidelines and keeping them relevant. This could be organized more efficiently at a central level than by the professional associations.

Guidelines can be a valuable instrument of evidence-based medicine. They translate scientific knowledge into practice, but they are no substitute for medical judgment. Physicians are not sufficiently trained to appraise new knowledge independently, and guidelines threaten to replace evidence-based medicine with rule-following medicine. Sackett stressed that external evidence can inform a physician, but can never replace one. The art of medicine is to determine whether and how research based on specific groups of patients applies to the unique patient now sitting in front of the doctor, not to apply a guideline to every patient.

Dasja Pajkrt
Professor of Pediatric Viral Infectious Diseases, University of Amsterdam

Sander van Deventer
Emeritus Professor of Gastroenterology and Hepatology, University of Amsterdam
Emeritus Professor of Translational Gastroenterology, Leiden University
Operating Partner, Forbion